

Dynamics:

$$\begin{aligned}\dot{x} &= V \cos(\theta) \\ \dot{y} &= V \sin(\theta) \\ \dot{\theta} &= w\end{aligned}$$

← ODE of state

Constraints:

$$\begin{aligned}(x_0, y_0, \theta_0) &= (0, 0, \frac{\pi}{2}) \\ (x_T, y_T, \theta_T) &= (5, 5, \frac{\pi}{2})\end{aligned}$$

Cost: $J = \int_0^T (\alpha + v(t)^2 + w(t)^2) dt$

want: $t^*, u^* = (v^*, w^*)$

$$H = g + pf$$

$$= \alpha + v(t)^2 + w(t)^2 + P_x V \cos(\theta) + P_y V \sin(\theta) + P_\theta w$$

$$\begin{cases} \dot{P}_x^* = -\frac{\partial H}{\partial x} = 0 \\ \dot{P}_y^* = -\frac{\partial H}{\partial y} = 0 \\ \dot{P}_\theta^* = -\frac{\partial H}{\partial \theta} = -(-P_x V \sin(\theta) + P_y V \cos(\theta)) \\ \quad = P_x V \sin(\theta) - P_y V \cos(\theta) \end{cases}$$

← ODE of co-state

$$\therefore \begin{cases} \frac{\partial H}{\partial v} = 2v + P_x \cos(\theta) + P_y \sin(\theta) = 0 \\ \frac{\partial H}{\partial w} = 2w + P_\theta = 0 \end{cases}$$

← optimal control

$$\therefore \begin{cases} v^* = -\frac{1}{2} (P_x^* \cos(\theta^*) + P_y^* \sin(\theta^*)) \\ w^* = -\frac{1}{2} P_\theta^* \end{cases} \Rightarrow \begin{cases} P_x \cos \theta + P_y \sin \theta = -2v \\ P_\theta = -2w \end{cases}$$

Free final time. Fixed final state

$$\therefore H(\tau) + \frac{\partial h}{\partial t} = 0$$

$$\therefore H(\tau) = 0$$

$$\therefore \alpha + V^2 + w^2 + p_x V \cos \theta + p_y V \sin \theta + p_\theta w = 0$$

$$\alpha + V^2 + w^2 + V \cdot (-2V) + (-2w) \cdot w = 0$$

$$\alpha + V^2 + w^2 - 2V^2 - 2w^2 = 0$$

$$\therefore \alpha = V(\tau)^2 + w(\tau)^2$$

$$= \frac{1}{4} [p_x(\tau) \cos(\theta) + p_y(\tau) \sin(\theta)]^2 + \frac{1}{4} [p_\theta(\tau)]^2$$

$$\therefore \theta = \frac{\pi}{2}$$

$$\therefore \cos \theta = 0, \sin \theta = 1$$

$$\therefore \alpha = \frac{1}{4} p_y(\tau)^2 + \frac{1}{4} p_\theta(\tau)^2$$

$$\therefore 4\alpha = p_y(\tau)^2 + p_\theta(\tau)^2$$

← boundary conditions